Zecpath HR Interview AI – System Architecture

Objective

This document describes the architecture of the HR Interview AI system, including core modules, data flow, and system interactions.

High-Level Architecture

Frontend (Web / Mobile App)

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Backend API Layer

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HR Interview AI System

- Question Generator
- Conversation Engine
- Follow-Up Engine
- Answer Understanding Engine
- Communication Analyzer
- Confidence & Behavior Analyzer
- HR Scoring Engine
- Aptitude Engine
- Summary Generator

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Database / Storage

Data Flow

User Input (Voice/Text)

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Speech-to-Text Processing

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Answer Processing

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AI Evaluation

- Intent Detection
- Communication Analysis
- Confidence Analysis
- Scoring

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Report Generation

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API Response

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Recruiter Dashboard

Core Modules

Module	Responsibility
Question Generator	Generates interview questions
Conversation Engine	Manages interview flow
Follow-Up Engine	Creates contextual follow-up questions
Answer Engine	Extracts intent and meaning
Communication Analyzer	Evaluates communication quality
Confidence Analyzer	Detects confidence signals
HR Scoring Engine	Calculates interview scores
Summary Generator	Creates recruiter-friendly reports

Storage Layer

Candidate Data:

- Resume
- Interview Responses
- Scores
- Reports

Database Options:

- PostgreSQL

- MongoDB

Technology Stack

Backend:

- Python
- Flask / FastAPI

AI Layer:

- NLP
- Rule-Based Models
- Machine Learning

Database:

- PostgreSQL
- MongoDB

System Benefits

- Modular architecture
- Easy maintenance
- Scalable design
- Explainable AI decisions

Future Improvements

- Real-time AI scoring
- Voice emotion analysis
- Multi-language support
- Streaming interview sessions